

Router vs. Switch Comparison Table

Feature	Router	Switch
Main Use	Connects different networks together (e.g., connecting your home network to the internet).	Connects multiple devices together within the same network to form a Local Area Network (LAN).
Typical Location	Positioned at the boundary of a network (the gateway connecting the home/office to the outside ISP line).	Positioned centrally within a local network setup (e.g., in an entertainment center, desk cluster, or server room cabinet).
How They Handle Data	Uses IP addresses (Network Layer) to route data packets intelligently between different networks.	Uses MAC addresses (Data Link Layer) to direct data frames specifically to the intended receiving device.
Data Transmission Type	Processes data in the form of Packets.	Processes data in the form of Frames.
Collision/Broadcast Domain	Creates separate broadcast domains and routing paths; manages traffic to prevent outside congestion.	Creates separate collision domains for each port, allowing devices to transmit data simultaneously without interference.